

COLLABORATOR SIDE



Collaborator X_j + LiDAR

PointPillars / BEV backbone
shared weights

Object-level message L
 $S = \{E, L, F\}; E = \text{ego-only, no}$
 $\approx 0.024 \text{ Msym/frame}$

Feature codec
Feature-level message F (16-QAM)
 $\approx 0.99 \text{ Msym/frame}$

C256: 256-QAM comparator (not deployed), $\approx 0.495 \text{ Msym/frame}$
dominated; excluded from deployment
(§III-B)

select one, controlled by s
(one message)

V2V wireless channel (SNR, BLER)

2-bit request (via CAM)

EGO SIDE — ON-VEHICLE DECISION



Ego vehicle X_e
"Selector runs on the ego vehicle"

LiDAR

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local detection output

Cue Extractor
 z_t

Channel estimator
(SNR, dB)
(AWGN or Rayleigh)

RF Selector
 $g(z_t, \hat{y}_t, \hat{\sigma}_t) \mapsto S_t$

Fusion Ψ
 $F_{ego,t} + m_{s_t}$

Detection head D

(3D boxes)

request: ego → collaborator
message: collaborator → ego